Tip-Guided Few-Shot Adaptation of Motion Time-Series Foundation Models

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Abstract

Cross-dataset few-shot human activity recognition (HAR) requires a pretrained motion encoder to classify activities on an unseen sensor dataset given only k labeled windows per class. Motion foundation models such as UniMTS supply strong zero-shot text alignment, yet two standard adaptation routes both fall short: full fine-tuning overfits sparse supports, while frozen Tip-Adapter caches saturate once domain shift exceeds what a fixed encoder can absorb. We introduce **Tip-Guided Adaptation (TipGuided)**, which fine-tunes the UniMTS ST-GCN encoder under an explicit Tip-cache prior that preserves CLIP-aligned text scores and support affinity structure. On the official UniMTS protocol—18 real HAR datasets, $k \in \{1, 2, 3, 5, 10\}$, leave-one-dataset-out Tip hyper-parameters—TipGuided raises mean accuracy by up to $+8.4$ points over frozen Tip-F and surpasses full fine-tuning at every k, with a peak margin of $+3.8$ points at $k = 2$. Training dynamics show TipGuided enters with higher test accuracy from Tip/text priors and fits the support objective faster; t-SNE embeddings show more class-separable target features after adaptation.

1 Introduction

Wearable IMU sensing powers healthcare monitoring, sports analytics, and mobile context awareness. Recent motion time-series foundation models, notably UniMTS [1], contrastively align a skeleton ST-GCN encoder with CLIP [3] text embeddings and report competitive zero-shot transfer across heterogeneous HAR corpora. Deployment on a new dataset still benefits from a handful of labeled windows per class. This defines a precise and practical problem: **cross-dataset few-shot HAR**.

Two adaptation families dominate this setting. Full fine-tuning updates the encoder and a linear head on the support set; it is strong, but prior-free optimization on $k \leq 10$ windows per class readily memorizes sensor-specific artifacts. Frozen alternatives keep the encoder fixed and learn a lightweight decision rule—linear probes, nearest neighbors, or Tip-Adapter [2] feature caches imported from CLIP few-shot classification. On UniMTS features, Tip-Adapter-F clearly outperforms probes and 1-NN. Further frozen refinements (soft values, late linear fusion, longer key training) add only ~ 0.5 points on average. The bottleneck is therefore not cache hyper-parameter search: **domain shift demands encoder updates that retain pretrained text–motion geometry**.

We propose **Tip-Guided Adaptation (TipGuided)**. TipGuided freezes the CLIP text tower and updates the ST-GCN encoder together with Tip keys and a CE head. Classification logits fuse three complementary signals (Figure 1):

$$z(x) = z_{\text{text}}(x) + \alpha\, A(x)V + \gamma\, W f_\theta(x), \tag{1}$$

where z_{text} is the zero-shot text score, $A(x)V$ is Tip affinity over support keys/values, and $W f_\theta(x)$ is a CE head on the updating embedding. The Tip term anchors predictions to support neighborhoods and language-aligned directions; the CE term exploits encoder plasticity.

Contributions.

1. We locate UniMTS transfer in official cross-dataset few-shot evaluation and show that frozen Tip-style adapters saturate under this protocol.
2. We introduce TipGuided, coupling Tip-cache priors with encoder fine-tuning for motion foundation models.
3. On 18 HAR datasets with official k-shot indices, TipGuided outperforms Tip-F and beats full fine-tuning for all $k \in \{1, 2, 3, 5, 10\}$.

2 Related Work

Motion / IMU foundation models. UniMTS [1] pretrains ST-GCN [4] over a canonical 22-joint layout and aligns embeddings with CLIP text for zero-shot HAR. Our work studies post-hoc few-shot adaptation under UniMTS's public protocol rather than replacing pretraining.

Few-shot adaptation of foundation models. Tip-Adapter and Tip-Adapter-F [2] build keys/values from support features and fuse cache logits with zero-shot CLIP scores. We transplant this mechanism to UniMTS features, then show that a frozen Tip cache is insufficient for cross-dataset IMU shift, motivating encoder-side TipGuided updates. Parameter-efficient fine-tuning (adapters, LoRA) and classical few-shot learners (ProtoNet [5]) share the same budgeted-adaptation goal; TipGuided specifically reuses Tip geometry already native to language-aligned motion encoders.

Fine-tuning under distribution shift. Full fine-tuning remains a strong HAR transfer baseline but erases pretrained structure when supports are tiny. TipGuided is fine-tuning with an explicit nonparametric prior constructed from the same supports.

3 Preliminaries

3.1 UniMTS encoder and text alignment

Let x denote a windowed IMU sequence mapped into a 22-joint skeleton tensor. UniMTS encodes $f(x) \in \mathbb{R}^d$ (d=512) with ST-GCN-18 and compares it to CLIP text embeddings of activity names. We use a four-template prompt ensemble. Zero-shot logits are $z_{\text{text}}(x) = \tau\, f(x)^\top T$ with temperature τ and stacked text matrix T.

3.2 Tip cache

Given a k-shot support set $\{(x_i, y_i)\}_{i=1}^N$, Tip keys are $K = [f(x_i)]$ and values V are class-balanced one-hot labels. For a query with ℓ_2-normalized feature f,

$$A_i = \exp\big(-\beta(1 - \langle f, k_i\rangle)\big), \qquad z_{\text{tip}} = \alpha\, AV.$$

Tip-Adapter-F optimizes keys (and optionally α) on the support set with the encoder frozen.

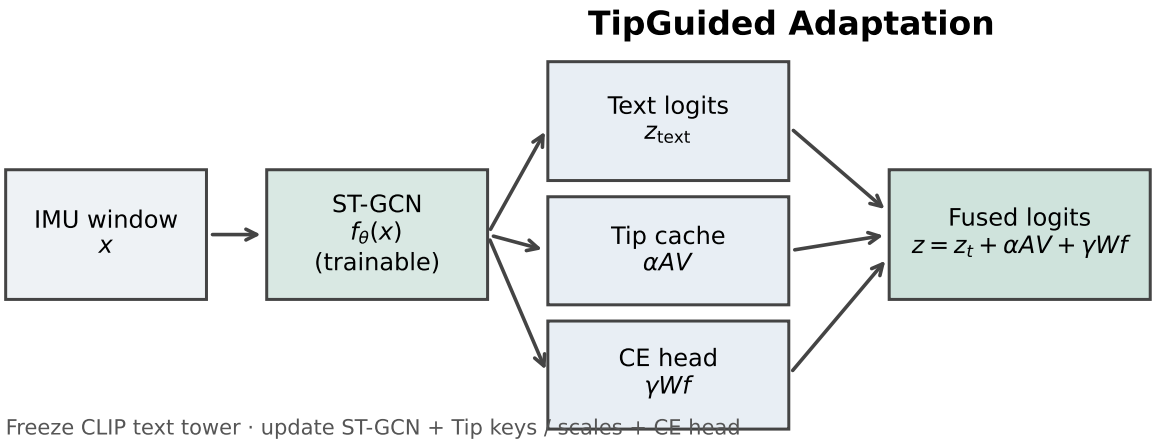


Figure 1: TipGuided. A trainable ST-GCN produces $f_\theta(x)$. Logits fuse text zero-shot scores, Tip-cache affinity over support keys/values, and a CE head. The CLIP text tower remains frozen.

3.3 Evaluation protocol

We follow UniMTS official few-shot files `few_shot_data_2/{dataset}_k={k}.pth` on 18 datasets: Opp_g, UCIHAR, MotionSense, w-HAR, Shoaib, har70plus, realworld, TNDA-HAR, PAMAP, USCHAD, Mhealth, Harth, ut-complex, Wharf, WISDM, DSADS, UTD-MHAD, MMAct. For Tip hyper-parameters (β, α), leave-one-dataset-out selection on the other 17 datasets' mean accuracy evaluates the held-out dataset. Primary metric: mean accuracy over the 18 datasets; we also report macro-F1 for TipGuided.

4 Method: Tip-Guided Adaptation

4.1 Objective

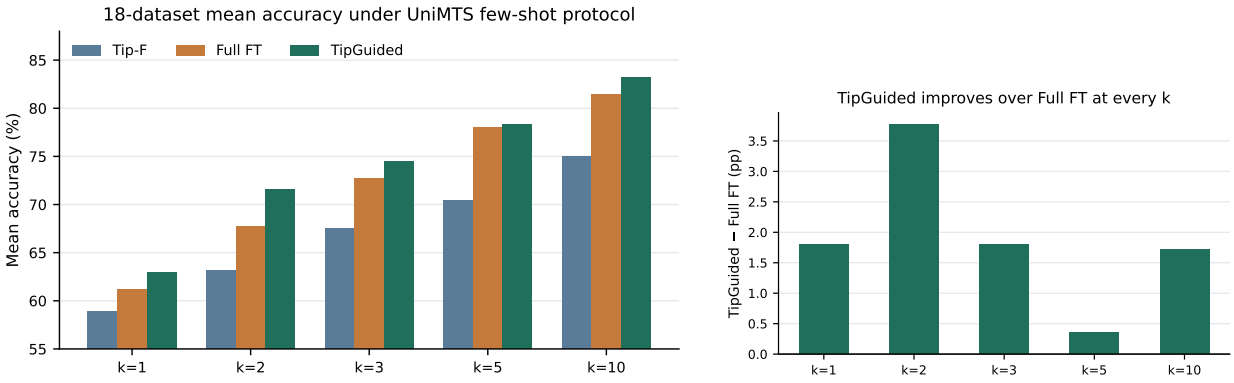
TipGuided initializes from a UniMTS checkpoint, freezes the CLIP text tower, and updates ST-GCN parameters θ together with Tip keys K, scales α, γ, and CE head W. On support samples we minimize cross-entropy of the fused logits in Eq. (1). Keys initialize from current support embeddings and optimize jointly; values remain support-derived and class-balanced.

4.2 What the prior contributes

Full FT optimizes only $W f_\theta(x)$. With tiny N, gradients push f_θ toward support memorization and away from language-aligned geometry. The Tip term keeps predictions consistent with support neighborhoods; z_{text} preserves semantic directions from pretraining. Together they act as a data-dependent regularizer rather than a separate classifier bolted on after adaptation.

4.3 Operating regimes

Ablations identify a clear recipe: **low shot** ($k \leq 3$) favors joint Tip+CE fusion; **mid/high shot** ($k \geq 5$) favors CE-primary training with Tip as initialization/regularizer under moderate learning rates ($\sim 5{\times}10^{-5}$) and 100–150 epochs.



(a) Main comparison.

(b) Gap vs Full FT.

Figure 2: Cross-dataset few-shot accuracy. TipGuided exceeds Tip-F and Full FT for all official k.

Table 1: 18-dataset mean accuracy on official UniMTS few-shot indices. TipGuided reports the best variant per k (fuse for $k \leq 3$; CE-primary for $k \geq 5$).

k	ZS	1-NN	Probe	Tip-F	Tip+	Full FT	**TipGuided**
1	0.444	0.578	0.458	0.590	0.599	0.612	**0.630**
2	0.444	0.630	0.530	0.632	0.646	0.678	**0.716**
3	0.444	0.650	0.546	0.676	0.673	0.728	**0.746**
5	0.444	0.686	0.519	0.705	0.709	0.780	**0.784**
10	0.444	0.706	0.615	0.751	0.752	0.815	**0.832**

5 Experiments

5.1 Baselines

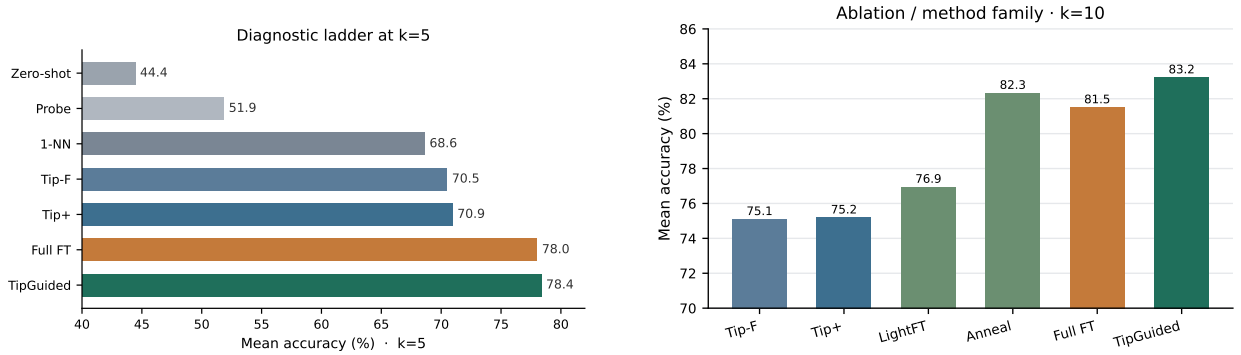
Zero-shot UniMTS; cosine 1-NN; linear probe on frozen features; Tip / Tip-F with LOO (β, α); Tip+ frozen refinements; LightFT (partial/full encoder FT with Tip); Full FT (encoder+CE, 100 epochs); TipGuided (fuse / CE-primary / anneal variants). Unless noted, TipGuided reports the best variant per k under the recipes above.

5.2 Main results

Figure 2 and Table 1 summarize 18-dataset mean accuracy. TipGuided is best at every k. Relative to Tip-F, gains range from $+4.0$ (k=1) to $+8.4$ (k=2) points. Relative to Full FT, all gaps are positive, peaking at $+3.8$ points for k=2.

5.3 Diagnostic ladder

Figure 3a orders methods at k=5. Zero-shot and linear probes sit far below Tip-F, confirming that cache adaptation exploits support geometry beyond a linear head. Tip+ barely moves Tip-F, marking frozen saturation. Full FT jumps into the high 0.78 range; TipGuided clears that bar.



(a) Method ladder at k=5.

(b) Method family at k=10.

Figure 3: Where gains come from. Frozen Tip saturates; encoder updates with Tip priors deliver the remaining lift.

5.4 Per-dataset behavior

Figure 4 compares Tip-F, Full FT, and TipGuided on each dataset. At k=10, TipGuided wins on $12/18$ datasets, with large gains on w-HAR, Wharf, ut-complex, MMAct, and realworld—settings where prior-free FT overfits heterogeneous or smaller targets. At k=5, TipGuided still trails Full FT on Shoaib, realworld, and USCHAD, while leading on Wharf, UTD-MHAD, and MMAct; the mean remains higher because wins concentrate on hard, low-resource targets.

5.5 Training dynamics

Figure 5 shows Tip-F key training: support CE drops quickly while test accuracy plateaus—the frozen encoder never moves. Figure 6 contrasts Full FT and TipGuided on matched supports and budgets. TipGuided starts from a much stronger test accuracy because Tip/text logits are available from epoch 0, and its fused objective drives support CE down faster than CE-only Full FT. On UCIHAR and PAMAP, TipGuided remains competitive throughout; on Wharf the short-horizon curve is noisier, consistent with that dataset's small test pool. The 18-dataset means in Table 1 aggregate these trajectories under the full training recipe.

5.6 Feature geometry

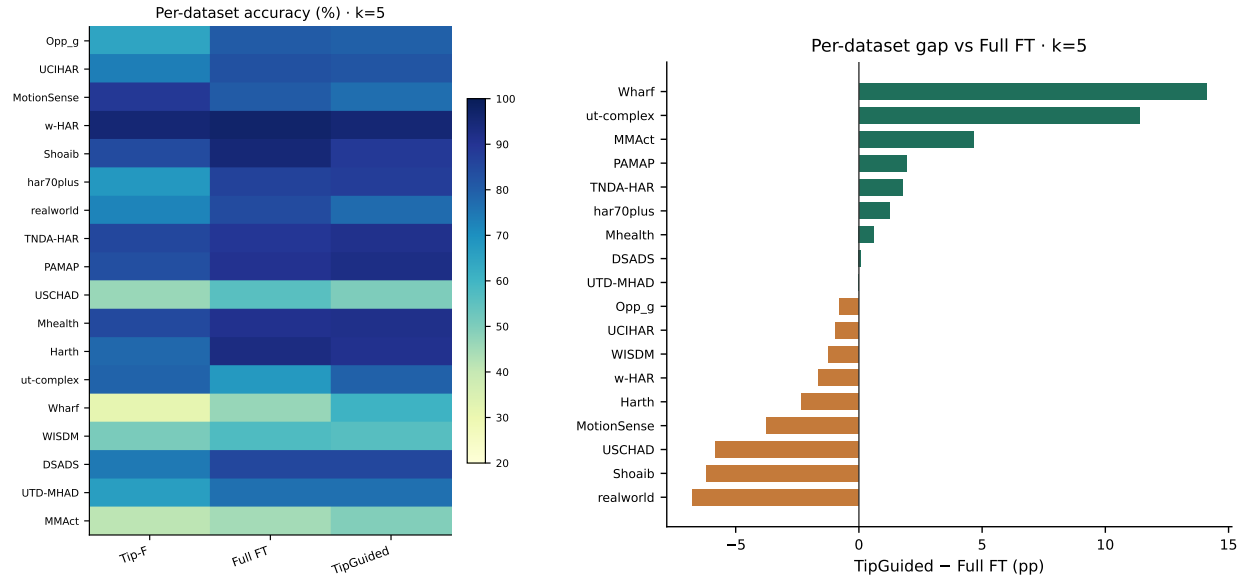
Figure 7 visualizes t-SNE of test embeddings before and after TipGuided-CE on representative datasets. After adaptation, class clusters tighten and inter-class gaps widen relative to the frozen UniMTS features, matching the accuracy gains in Table 1.

5.7 Ablations

At k=10 (Figure 3b), LightFT and Tip→CE annealing improve over Tip-F yet remain below TipGuided CE (0.832). Longer aggressive schedules without Tip structure do not substitute for the prior: a longer CE-only TipGuided run at k=10 underperforms the default 100-epoch recipe.

6 Discussion

TipGuided reframes Tip-Adapter from a frozen classifier into an adaptation prior for encoder fine-tuning. The central empirical fact is simple: under UniMTS's cross-dataset few-shot protocol, keeping Tip/text geometry



(a) Per-dataset accuracy, k=5.

(b) TipGuided − Full FT, k=5.

Figure 4: Dataset-level view. TipGuided concentrates gains where Full FT overfits.

while updating the motion encoder beats discarding that geometry (Full FT) and beats refusing to update the encoder (Tip-F).

The k-dependent recipe is itself informative. When supports are extremely sparse, Tip logits supply supervision the CE head cannot yet estimate—hence the large early-epoch gap in Figure 6. When supports grow, CE-primary decoding dominates and Tip mainly shapes the adaptation trajectory.

Scope. All numbers use UniMTS checkpoints and official splits. Extending TipGuided to other IMU foundation models, confirming the tight k=5 margin with multiple seeds, and collapsing fuse/CE-primary into one tip-weight schedule are concrete follow-ups.

7 Conclusion

Cross-dataset few-shot HAR exposes a clear failure mode of both frozen Tip caches and prior-free full fine-tuning. TipGuided closes that gap by fine-tuning UniMTS under Tip-cache and text priors, improving mean accuracy over Full FT for every official k on 18 datasets. Adaptation should update the encoder while keeping foundation-model geometry explicit.

Acknowledgments

[Anonymous.]

References

[1] Yan et al. UniMTS: Unified Pre-training for Motion Time Series. In *NeurIPS*, 2024.

[2] Zhang et al. Tip-Adapter: Training-free Adaption of CLIP for Few-shot Classification. In *ECCV*, 2022.

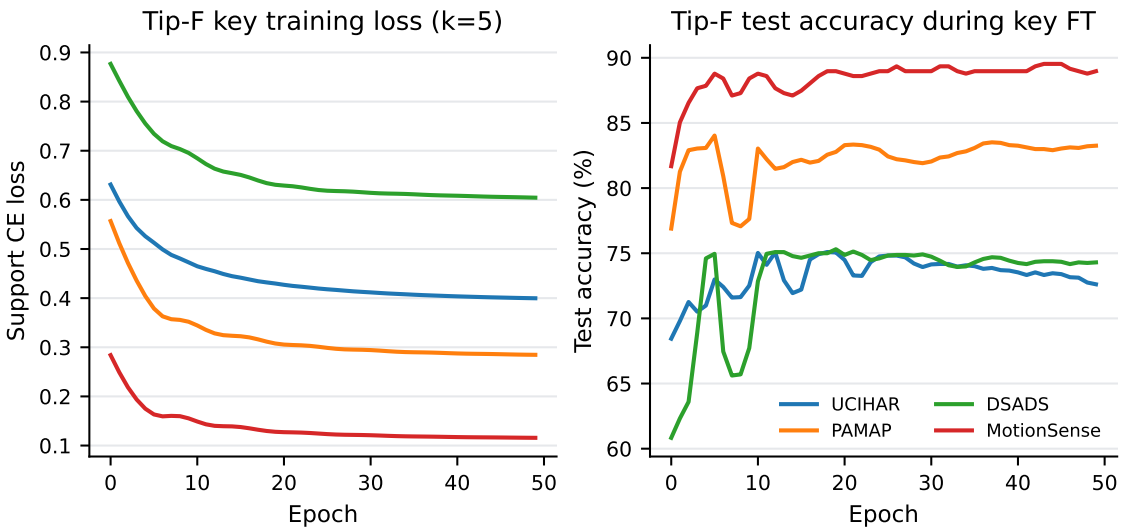


Figure 5: Frozen Tip-F dynamics (k=5). Key fine-tuning fits the support set, yet test accuracy saturates—the encoder never moves.

[3] Radford et al. Learning Transferable Visual Models From Natural Language Supervision. In *ICML*, 2021.

[4] Yan, Xiong, and Lin. Spatial Temporal Graph Convolutional Networks for Skeleton-Based Action Recognition. In *AAAI*, 2018.

[5] Snell, Swersky, and Zemel. Prototypical Networks for Few-shot Learning. In *NeurIPS*, 2017.

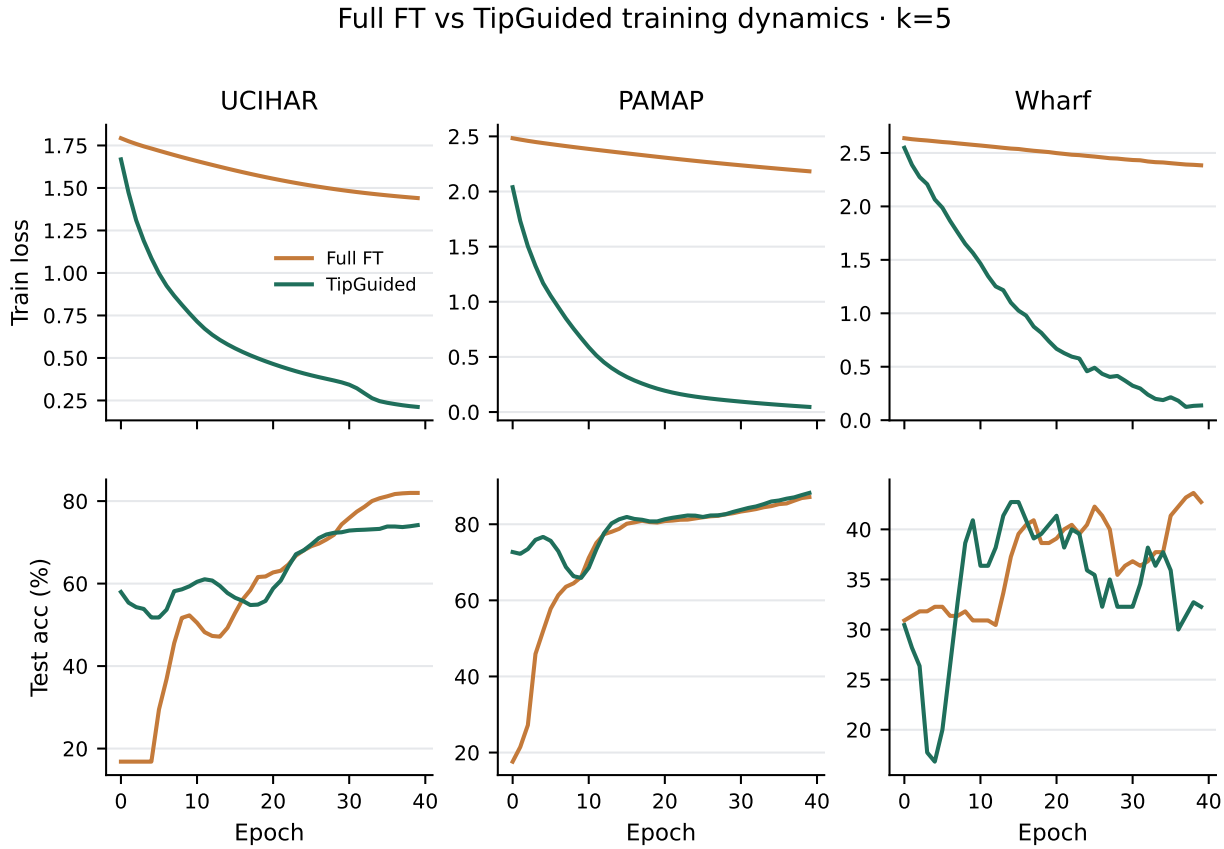


Figure 6: Full FT vs TipGuided training (k=5, 40 epochs). TipGuided enters with higher test accuracy from Tip/text priors and reduces support CE faster via the fused objective.

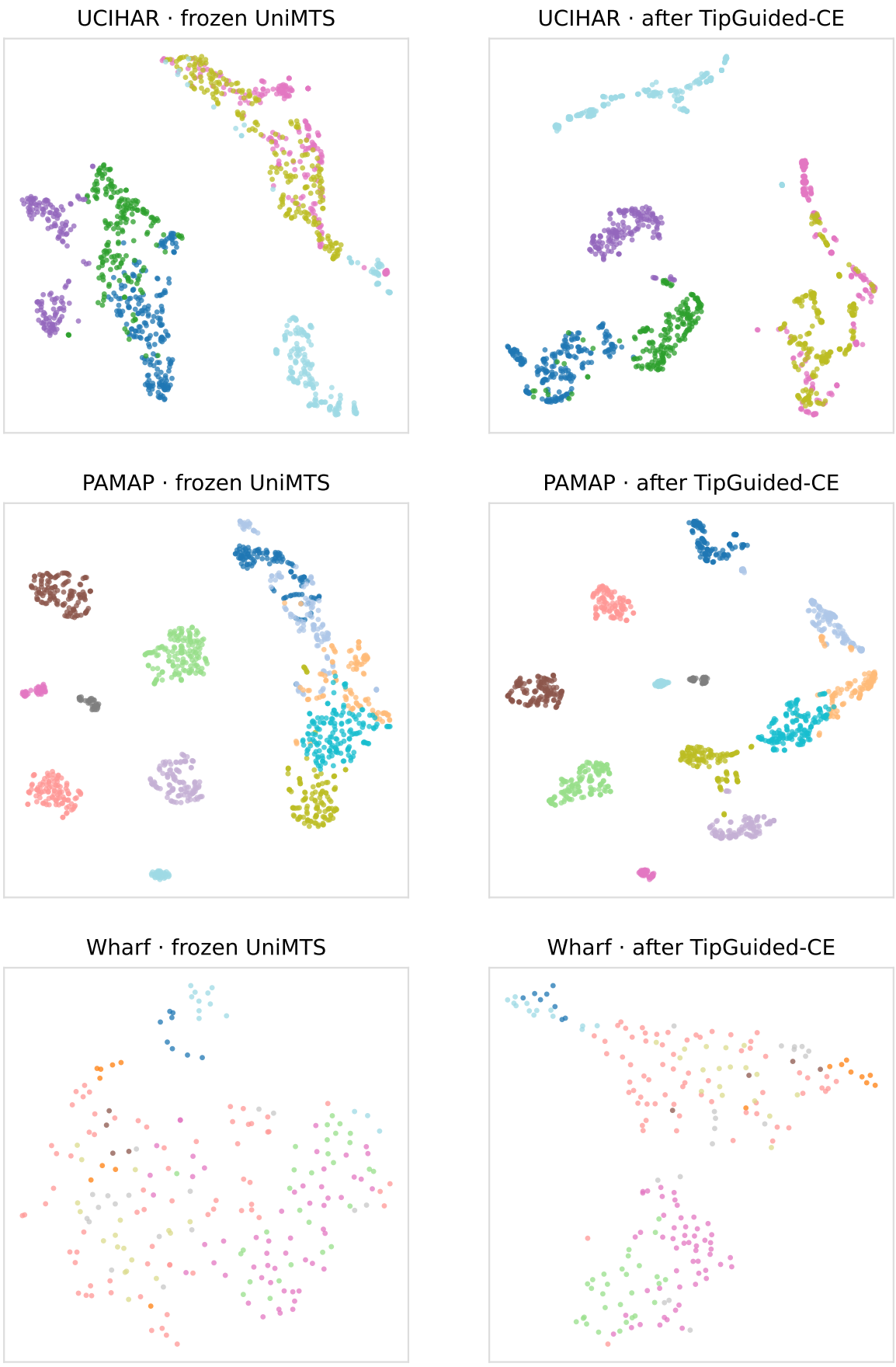


Figure 7: t-SNE of test features (k=5). Each row: frozen UniMTS (left) vs TipGuided-CE (right) for UCIHAR, PAMAP, and Wharf. TipGuided tightens class clusters and widens inter-class gaps.